The Curious Case of SignMuon: To Error Feedback or not to Error Feedback?

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Abstract

The SignSGD gradient compression algorithm enables up to a $32\times$ reduction in communication volume, which is critical for federated learning in bandwidth-constrained environments. However, it often lags behind modern optimizers like Muon, which leverage the matrix structure of parameters to achieve superior convergence and performance. In this work, we propose SignMuon, an algorithm that applies sign compression to the Linear Minimization Oracle (LMO) update of the Muon optimizer. Our empirical results demonstrate that SignMuon achieves accuracy nearly on par with Muon while significantly outperforming SignSGD in both centralized and federated learning settings.

Keywords: federated learning, stochastic optimization, gradient compression, 1-bit quantization, matrix orthogonalization, Linear Minimization Oracle (LMO), SignSGD, Muon.

1 Introduction

Modern deep neural network (DNN) training methods require optimizers that not only provide high convergence rates but also minimize communication overhead in distributed systems. In federated learning, data are distributed across clients that communicate with a central server to train a global model. Each client possesses its own datasets and updates its parameters using a local optimizer. The goal of federated learning is to minimize the averaged loss function across all clients. Under limited network bandwidth, transmitting full model updates may significantly slow down the training process. Gradient compression methods are typically used to address this issue. Bernstein et al. (2018; 2019); Beznosikov et al. (2024)

For instance, SignSGD algorithm has demonstrated that even extreme compression (down to 1 bit per component) preserves the practical effectiveness of stochastic optimization and reduces the communication volume. This data volume is reduced by a factor of approximately $32\times$ compared to standard SGD, with only minor degradation in model quality Bernstein et al. (2018; 2019). However, SignSGD does not account for the matrix structure of the parameters. In tasks with matrix weights, it treats matrices as one-dimensional vectors (flattening), ignoring their intrinsic two-dimensional tensor structure, which may negatively affect the training performance of DNN.

Recently, Muon has demonstrated that orthonormalized updates can stabilize DNN training and, in several cases, outperform standard adaptive optimization methods in terms of final solution accuracy Jordan et al. (2024); Bernstein and Newhouse (2024); Essential AI (2025). For matrix parameters, Muon implements a steepest descent step in the spectral norm. According to Jordan et al. (2024), the update direction is computed as the orthonormal matrix $\mathbf{U}\mathbf{V}^\top$, where $\mathbf{U}$ and $\mathbf{V}$ are the singular-vector matrices obtained from the gradient decomposition. Based on the structure of the algorithm, Muon requires the transmission of full update matrices, which can be prohibitively computationally expensive for federated tasks.

In this work, we propose SignMuon[1], an optimization algorithm that applies sign compression to the Linear Minimization Oracle (LMO) update of Muon. SignSGD transmits the sign of the coordinates of the stochastic gradient direction. Our algorithm first computes an update direction using LMO, and then transmits only the sign of this direction. This approach allows for the matrix structure of the parameters to be accounted for and preserves the advantages of Muon, while reducing the amount

[1] Code is available at: `https://github.com/intsystems/2026-Project-203/tree/main`

of transmitted information to one bit per coordinate. Consequently, SignMuon combines the fast convergence of structured optimizers with the communication efficiency of sign-based approaches.

The effectiveness of the proposed optimizer is evaluated on synthetic and practical tasks: centralized image classification on CIFAR-10 with a ResNet-18 backbone, federated image classification on CIFAR-10 with a CNN across several client counts, and on a synthetic smooth convex least-squares problem. Our experiments investigate how effectively Muon-style structured update directions can be combined with the extreme communication compression characteristic of sign-based methods. Specifically, we analyze how transmitting only the update direction signal affects the accuracy and stability of training compared to using the full Muon update.

Experimental results show that applying sign compression to Muon directions reduces communication overhead (by $32\times$) while maintaining optimization rates. In the centralized setting, SignMuon consistently outperforms the baseline SignSGD method in both convergence speed and final accuracy. At the same time, the accuracy of the proposed 1-bit method is only $1 - 1.5\%$ lower than that of the Muon optimizer, while significantly outperforming basic signed algorithms. In federated setting, integrating SignMuon with client momentum and Majority Vote aggregation method provides stable convergence and robustness to anomalous client updates even under severe communication constraints. Furthermore, we present an analytical example that formalizes the limitations of applying sign compression to LMO-based updates in the deterministic setting and for smooth convex optimization problems. The results of our experiments demonstrate that the proposed algorithm successfully combines LMO-based methods with sign compression without any significant loss in training accuracy.

2 Related Work

3 Problem Statement

Centralized Learning: We consider the stochastic optimization problem:

$$\min_{\mathbf{X}\in\mathcal{X}} \{f(\mathbf{X}) := \mathbb{E}_{\xi\sim\mathcal{D}}[f_\xi(\mathbf{X})]\}, \tag{1}$$

where $\mathcal{X}$ is the parameter space (e.g., $\mathbb{R}^d$ or $\mathbb{R}^{m\times n}$), functions $f_\xi : \mathcal{X} \to \mathbb{R}$ are potentially non-convex and non-smooth but continuously differentiable functions. Here, $f_\xi(\mathbf{X})$ is the loss function of model parameterized by $\mathbf{X}$, computed at a data point ξ, sampled from the probability distribution $\mathcal{D}$.

Federated Learning: In modern machine learning, progress is increasingly driven by the ability to scale computational systems. State-of-the-art models rely on massive datasets and highly complex architectures, often requiring substantial computational resources and extended training times. As a result, the training process is formulated as an optimization problem in a distributed environment:

$$\min_{\mathbf{X}\in\mathcal{X}} \left\{ f(\mathbf{X}) := \frac{1}{N}\sum_{j=1}^{N} f_j(\mathbf{X}) \right\}, \quad f_j(\mathbf{X}) = \mathbb{E}_{\xi_j\sim\mathcal{D}_j}[f_j(\mathbf{X}, \xi_j)] \tag{2}$$

where $\mathcal{X}$ is the parameter space (e.g.,$\mathbb{R}^d$ or $\mathbb{R}^{m\times n}$), $\mathbf{X} \in \mathcal{X}$ are the model parameters, $N \geq 1$ is the number of clients, and $f_j(\mathbf{X})$ is the loss of the model $\mathbf{X}$ on the data $\mathcal{D}_j$, stored on client $j \in [N] := \{1, \dots, N\}$. In federated learning, clients do local computations and synchronize with a global server. Since communication bandwidth is often limited, the volume of transmitted data becomes a major bottleneck in the training process Bernstein et al. (2019); Horváth et al. (2019). Several practical techniques for federated optimization and communication compression have also been studied in Beznosikov et al. (2024); Gruntkowska et al. (2025); Ahn et al. (2025).

For theoretical optimization analysis in distributed environments, we rely on Assumption 2 that the objective function f is smooth w.r.t. the spectral norm. We note that, in practice, weaker smoothness assumptions such as the generalized (L_0, L_1)-smoothness condition may also be considered Pethick et al. (2026). Limited network bandwidth makes the transmission of full-precision 32-bit gradients prohibitively expensive, especially when training large-scale models. This motivates the study of communication-efficient optimization methods based on extreme gradient compression, where the communication budget is reduced to a single bit per transmitted coordinate Bernstein et al. (2018; 2019); Beznosikov et al. (2024).

Linear Minimization Oracle (LMO) is defined as an operator $A(\cdot)$ that provides the solution to a linear minimization problem over a feasible set:

$$A(\mathbf{X}) = \arg\min_{\mathbf{Y}\in\mathcal{S}} \langle \mathbf{X}, \mathbf{Y} \rangle. \tag{3}$$

where $\mathbf{X} \in \mathbb{R}^{m\times n}$ is the input matrix, $\mathcal{S} \subset \mathbb{R}^{m\times n}$ is the convex set, and $\langle \mathbf{X}, \mathbf{Y} \rangle = \sum_{i,j} X_{ij}Y_{ij}$ is the standard matrix inner product Pethick et al. (2025). In particular, the constraints can be defined via the unit ball of a given norm:

$$A(\mathbf{X}) = \arg\min_{\mathbf{Y}\in\{\mathbf{Y}\in\mathcal{S}|\|\mathbf{Y}\|\leq 1\}} \langle \mathbf{X}, \mathbf{Y} \rangle. \tag{4}$$

Such an operator selects the direction of steepest descent with respect to the specified norm. The analytical and practical role of the LMO is discussed in detail in the literature on norm-constrained LMO optimizers Pethick et al. (2025); Kovalev (2025); Riabinin et al. (2025).

Muon optimizer uses the matrix structure of the gradient. Muon implements a steepest descent step in the spectral norm, where the update direction is obtained by orthogonalizing the gradient matrix Bernstein and Newhouse (2024). Let $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the singular value decomposition (SVD) of the matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$, where $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ are the orthonormal matrices of singular vectors, $\mathbf{\Sigma} \in \mathbb{R}^{r\times r}$ is the diagonal matrix of singular values, and $r = \mathrm{rank}(\mathbf{M})$. Then, the LMO direction takes the following form:

$$A(\mathbf{M}) = \mathbf{U}\mathbf{V}^\top \in \mathbb{R}^{m\times n}. \tag{5}$$

That is, Muon selects an orthonormal update direction corresponding to the solution of the linear minimization problem induced by the spectral norm geometry. In practice, Muon utilizes a smoothed gradient estimate with momentum, and its update is defined as follows:

$$\begin{aligned}
\mathbf{M}_t &= \mu\mathbf{M}_{t-1} + \mathbf{G}_t, \\
\mathbf{M}_t &= \mathbf{U}_t\mathbf{\Sigma}_t\mathbf{V}_t^\top, \\
\mathbf{D}_t &= -A(\tilde{\mathbf{M}}_t) = \mathbf{U}_t\mathbf{V}_t^\top \\
\mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t\mathbf{D}_t.
\end{aligned} \tag{6}$$

In the original Muon implementation, the computation of $\mathbf{U}_t\mathbf{V}_t^\top$ is approximated using Newton-Schulz iterations and related polynomial schemes. It enables efficient matrix orthogonalization without an explicit SVD computation Amsel et al. (2026); Li and Hong (2025); Liu et al. (2025). Such orthogonalization produces more isotropic updates and prevents a small number of dominant singular directions from controlling the optimization process, which is particularly beneficial for training DNN Bernstein and Newhouse (2024); Essential AI (2025).

Despite its theoretical and practical advantages, Muon requires transmitting the full orthogonalized update matrix $\mathbf{D}_t$. This step makes Muon more expensive in federated learning settings. Our objective is to develop a communication-efficient optimization algorithm, that preserves the geometric properties of Muon's LMO-based updates while reducing the communication cost to the level of SignSGD (1 bit per parameter). At the same time, the proposed method should retain the standard convergence guarantees for smooth non-convex optimization:

$$\min_{0\leq t\leq T} \mathbb{E}\big[\|\nabla f(\mathbf{X}_t)\|_*^2\big] = \mathcal{O}(T^{-1/2}). \tag{7}$$

4 Theory

Consider the SignA family of algorithms, where A denotes any optimizer based on LMO directions. All methods in this family follow the same three-stage structure: momentum accumulation, computation of an LMO direction, and sign-compression of the resulting update.

Formally, a SignA method initializes the momentum buffer as $\mathbf{M}_0 = \mathbf{0}$, and at each iteration $t \geq 1$ performs the following sequence of updates:

$$\begin{aligned}
\mathbf{M}_t &= \mu \mathbf{M}_{t-1} + \mathbf{G}_t && \text{(Momentum accumulation)} \\
\tilde{\mathbf{M}}_t &= \begin{cases} \mathbf{G}_t + \mu \mathbf{M}_t & \text{(Nesterov momentum)}, \\ \mathbf{M}_t & \text{(otherwise)} \end{cases} && \text{(Effective direction)}, \\
\mathbf{D}_t &= -A(\tilde{\mathbf{M}}_t), && \text{(LMO direction)} \\
\mathbf{s}_t &= \operatorname{sign}(\mathbf{D}_t) \in \{\pm 1\}^{m \times n}, && \text{(Sign compression)} \\
\mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t \lambda_{\text{mult}} \mathbf{s}_t && \text{(Parameter update)},
\end{aligned} \tag{8}$$

where $\mathbf{X}_t \in \mathbb{R}^{m \times n}$ is the current model parameter matrix, $\mathbf{G}_t$ is its stochastic gradient, $\mu \in [0, 1)$ is the momentum coefficient, η_t is the learning rate, and $\lambda_{\text{mult}} > 0$ is an optional scaling factor (with $\lambda_{\text{mult}} = 1$ by default). The operator $A(\cdot)$ denotes a Linear Minimization Oracle (LMO) in a specified non-Euclidean norm, and the function $\operatorname{sign}(\cdot)$ is applied element-wise to the resulting structured direction $\mathbf{D}_t$.

The proposed SignMuon algorithm is a particular instance of the SignA family. It combines the orthogonalized LMO directions of the Muon optimizer with the extreme sign-based compression strategy of SignSGD. In the following sections, we describe two main variants of SignMuon: a centralized version and a federated version.

4.1 Centralized Learning

In the centralized setting, SignMuon is applied to the matrix-valued parameters of a neural network. The practical implementation takes into account the tensor structure of model parameters. The optimization process is divided into two separate streams: matrix parameters (weights of convolutional and fully connected layers, $n_{dim} \geq 2$) are updated using the SignMuon LMO-based procedure, whereas vector parameters (bias, Batch Normalization layer, $n_{dim} = 1$) as well as the final classification layer are optimized using the standard AdamW optimizer. This design was originally introduced in the empirical description of Muon by Jordan et al. (2024). The first theoretical convergence result was derived by Li and Hong, who analyzed the smooth non-convex setting Li and Hong (2025) and subsequently formalized within the Gluon framework Riabinin et al. (2025).

For matrix parameters, the key component of SignMuon is the computation of an orthonormal LMO direction $\mathbf{D}_t \approx \mathbf{U}_t \mathbf{V}_t^\top$. To avoid the computational cost of performing a full singular value decomposition (SVD) at every optimization step, the orthogonalization is approximated using a 5th-order Newton-Schulz iteration. In an implementation, the Newton-Schulz coefficients are chosen as $a = 3.4445, b = 4.7750, c = 2.0315$. The main difference between SignMuon and the classical SignSGD algorithm lies in the object being compressed. In SignSGD, sign compression is applied directly to the stochastic gradient. In contrast, SignMuon first constructs a structured update direction $\mathbf{D}_t$, corresponding to the Muon LMO step and only then applies the elementwise sign operator to obtain $\operatorname{sign}(\mathbf{D}_t)$. Consequently, SignMuon preserves the geometry of orthonormal updates while reducing the transmitted information to one bit per component. The complete update procedure for matrix parameters is presented in Appendix A.1.

4.2 Federated Learning

In federated learning, the training data are distributed across N clients. Each client j possesses a local dataset $\mathcal{D}_j$ and computes a local loss function $f_j(\mathbf{X})$. The goal of the training process is to minimize the averaged loss function across all clients (2). Since clients do not exchange raw data and communicate only through messages sent to the server, communication efficiency becomes a critical concern in federated optimization.

In federated version of SignMuon, the server acts as an aggregation node, while the main optimization procedure is performed locally on the clients. At the beginning of communication round t, each client j receives a copy of the current global model. Rather than performing local parameter updates, the client computes an averaged stochastic gradient by accumulating gradients over n_steps local mini-batches. This gradient accumulation procedure reduces the variance of the stochastic gradient estimate before the application of the LMO operator. Formally, the accumulated gradient is defined as:

$$\mathbf{G}_t^{(j)} \leftarrow \frac{1}{n_{\text{steps}}} \sum_{i=1}^{n_{\text{steps}}} \nabla f_j(\mathbf{X}_t; \xi_{t,i}^{(j)}).$$

In accordance with the general SignA framework (8), each client maintains its own local momentum buffer $\mathbf{M}_t^{(j)}$. After computing the accumulated gradient, the client updates the momentum estimate, applies the LMO (implemented via Newton-Schulz orthogonalization), and then performs element-wise sign compression:

$$\mathbf{M}_t^{(j)} = \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$$
$$\mathbf{s}_t^{(j)} = \operatorname{sign}\big(A(\mathbf{M}_t^{(j)})\big) \in \{\pm 1\}^{m \times n}$$

where $t \in [0, \text{rounds} - 1]$, $A(\cdot)$ denotes the Muon operator implemented via Newton-Schulz orthogonalization with coefficients $a = 3.4445, b = 4.7750, c = 2.0315$. Thus, each client transmits only 1 bit per matrix parameter component to the server.

On the server, messages received from the client are aggregated using a majority vote. The aggregated sign is denoted by $\mathbf{s}_t^{\text{agg}}$, then it is given by the formula:

$$\mathbf{s}_t^{\text{agg}} = \operatorname{sign}\left(\sum_{j=1}^{N} \mathbf{s}_t^{(j)}\right). \tag{9}$$

The majority-vote operation ensures that the resulting sign $\mathbf{s}_t^{\text{agg}}$ is equal to $+1$ or -1 in each component and corresponds to the "majority vote" of the clients. Since momentum has already been taken into account at the clients' level, the server directly updates the global model of matrix parameters:

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{s}_t^{\text{agg}}.$$

A special treatment is applied to the final classification layer of the model. These parameters are not updated using the SignMuon rule, instead they are optimized using AdamW. The complete federated SignMuon procedure is presented in Appendix A.2.

4.3 Federated Learning with Error Feedback

Although Federated SignMuon demonstrates strong empirical stability, sign-based compression inevitably introduces bias into the descent direction. To compensate for quantization error and ensure strong convergence guarantees, we incorporate an Error Feedback method into our approach, adapted for LMO optimizers as described in a recent paper by Gruntkowska et al. Gruntkowska et al. (2025).

In this Federated EF-SignMuon variant, the orthogonalization procedure (LMO) is moved from the clients to the server. Instead of computing the LMO direction locally, each client evaluates the difference between its current local momentum estimate and the previous gradient estimate, and then applies sign compression to this residual. The server aggregates the compressed updates, reconstructs a global gradient estimate, and only then applies the Muon operator. The complete algorithm is presented in Appendix A.3.

The transmission of an additional scalar $\alpha_t^{(j)}$ for each layer introduces insignificant communication overhead, keeping the overall communication cost at ≈ 1 bit per parameter. Thus, both proposed federated SignMuon architectures: the baseline majority-vote scheme and the error-feedback variant – effectively address the communication bottleneck in federated learning. They preserve the geometric benefits of Muon-style matrix orthogonalization while maintaining an extremely low communication budget.

4.4 Theoretical limitations

The hypotheses presented in Appendix A.4 rely on the assumption that the direction of steepest descent in the spectral norm (the LMO step of the Muon algorithm) remains a valid descent direction even after applying extreme sign compression. However, this is not always the case.

The classic gradient descent algorithm guarantees that anti-gradient direction $(-\mathbf{G}_t)$ is a descent direction. In the SignMuon algorithm, the step direction is defined by the matrix $\mathbf{s}_t = \operatorname{sign}(\mathbf{U}_t \mathbf{V}_t^\top)$.

For the algorithm to guarantee convergence, the following descent condition must be met:

$$\langle \mathbf{G}_t, \operatorname{sign}(\mathbf{U}_t \mathbf{V}_t^\top) \rangle > 0. \tag{10}$$

This raises the question: does spectral orthogonalization guarantee that condition (10) holds for arbitrary matrices? We formalize this question in the following theorem.

Theorem 1. *There exist a function $f(\mathbf{W})$ and a matrix dimension $N \geq 4$ such that the SignMuon algorithm diverges.*

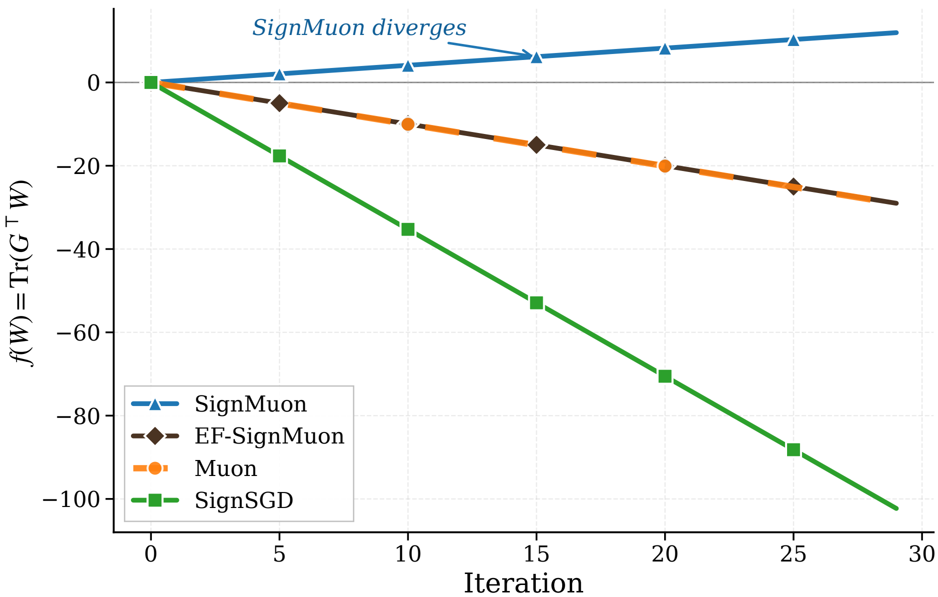

Figure 1: Comparison of optimization trajectories for the linear objective function $f(\mathbf{W}) = \mathrm{Tr}(\mathbf{G}^\top \mathbf{W})$, demonstrating the strict divergence of SignMuon, while other methods successfully minimize the function.

A proof by counterexample is provided in Appendix A.6. The existence of such an example demonstrates that SignMuon does not provide any global convergence for general smooth functions. Therefore, to explain its successful empirical performance on practical deep learning problems, where the algorithm achieves high accuracy, it is necessary to introduce additional assumptions.

5 Experiments

The goal of these experiments is empirical validation of algorithm SignMuon as an optimizer that combines the fast convergence properties of LMO-based methods with the communication efficiency of sign compression. The study consists of several parts, including evaluation in both centralized and federated settings, as well as validation on a synthetic least-squares optimization problem.

5.1 Smooth Convex Optimization Problem (Synthetic Experiment)

To evaluate the effect of matrix structure on the optimization process (without the influence of stochastic noise or the complexity of DNN architectures), we begin our experiments with a deterministic L-smooth convex optimization problem. We consider the minimization of a quadratic form:

$$F(\mathbf{X}) = \frac{1}{2}\|\mathbf{A}^{1/2}\mathbf{X}\mathbf{B}^{1/2}\|_F^2 = \frac{1}{2}\langle \mathbf{X}, \mathbf{A}\mathbf{X}\mathbf{B} \rangle \to \min_{\mathbf{X} \in \mathbb{R}^{m \times n}}, \tag{11}$$

where the parameter matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$ has dimension $m = n = 500$, which is representative of the size of parameter matrices in deep neural networks. The matrices $\mathbf{A} \in \mathbb{S}_+^m$ and $\mathbf{B} \in \mathbb{S}_+^n$ are symmetric positive semidefinite. To generate the problem instance, we randomly construct the spectra of $\mathbf{A}$ and $\mathbf{B}$ by sampling their eigenvalues uniformly from the interval $(0, 1)$, which guarantees a smoothness constant $L \leq 1$ with respect to the Frobenius norm. The parameter matrix $\mathbf{X}_0$ is initialized with elements sampled independently from a normal distribution $\mathcal{N}(0, 0.01)$.

Since theoretical convergence guarantees depend on the choice of norm and the corresponding smoothness constants, we adopt an empirical evaluation criterion. Specifically, the optimizers are compared by the minimum number of iterations required to reach the target function $F(\mathbf{X}) \leq 0.001$, the maximum number of iterations set to $T_{\max} = 5000$. We perform a grid search over the following hyperparameter ranges for each of the optimization algorithms under consideration:

Table 1: Hyperparameter search grid for the synthetic convex problem.

Algorithm	**Learning Rate** (η)	**Momentum** (μ)
SignMuon	$[1 \cdot 10^{-4}, 1 \cdot 10^{-3}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
Muon	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-3}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
EF-SignMuon	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-3}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
SignSGD	$[5 \cdot 10^{-5}, 2 \cdot 10^{-4}]$ with step $1 \cdot 10^{-5}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
SGD	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
Adam	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	—

The results of the grid search with the number of iterations required to converge to 0.001, are summarized in Table 2. Muon reaches the optimum in 1122 iterations. This is comparable to classical first-order methods (Adam, SGD). SignMuon reaches the target value in 2104 iterations, outperforming the baseline SignSGD algorithm, which requires 3120 iterations. The optimization trajectories in terms of both the objective value and the gradient norm are presented in Appendix A.7, Figure 2.

Table 2: Tuned learning rates and momentum coefficients in the experiments on the smooth convex optimization problem (Equation 11).

Algorithm	**Learning rate** (η)	**Momentum** (μ)	**Iterations to** 0.001
SignMuon	0.0002	0.2	2104
Muon	0.0065	0.1	1122
EF-SignMuon	-	-	-
SignSGD	0.00015	0.8	3120
SGD	0.1	0.95	972
Adam	0.07	-	202

5.2 Centralized Learning

In this experiment, we compare the convergence dynamics and the final accuracies of SignMuon with those of the classical optimizers: Muon, SignSGD, SGD, and Adam at a fixed number of epochs. One of the goals of the experiment is to demonstrate that SignMuon achieves classification quality comparable to Muon, while outperforming the classical SignSGD algorithm under the same communication budget (1 bit per parameter).

The experiments are conducted on CIFAR-10. It is a standard image classification dataset. It contains 60,000 color images of 10 classes. Each image is represented as a tensor of size $3 \times 32 \times 32$, where the first dimension corresponds to the number of color channels (RGB). For centralized training, the dataset is split into training and test sets. In the federated learning, the training set is partitioned across the clients.

The base model is a ResNet-18 architecture adapted to low-resolution images by using an initial 3×3 convolution without max-pooling. The network consists of 4 residual blocks with Batch Normalization (BatchNorm), followed by a dropout regularization layer and a linear classifier. As mentioned above, training is run for a fixed number of epochs $T_{\max} = 75$. During training, the learning rate is annealed with a cosine scheduler (CosineAnnealingLR). The learning rate η_t at epoch t is updated as follows:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{t}{T_{\max}}\pi\right)\right), \tag{12}$$

where $\eta_{\max}$ is the initial learning rate of the algorithm, $\eta_{\min} = 0$ is the minimum value (in our configuration), and $T_{\max}$ is the total number of training epochs.

The experimental results and the selected hyperparameter values are reported in Table 3. The dynamics of the objective function and of the accuracy of the algorithms during training are illustrated in Appendix A.8.

Table 3: CIFAR-10: centralized learning

Dataset	Optimizer	Epoch	Batch	LR	LR Aux	Mom	Train Acc	Test Acc
CIFAR-10	SignMuon	75	128	0.001	0.0001	0.9	99.69%	92.55%
	Muon			0.015	0.001		99.94%	93.70%
	EF-SignMuon						-%	-%
	SGD						99.98%	93.93%
	SignSGD			0.001	0.0001		93.19%	89.11%
	Adam						99.77%	93.03%

5.3 Federated Learning

The next stage of the research was adaptation SignMuon in the federated learning, where transmitting full 32-bit gradient matrices is the communication bottleneck. The goal of this stage is to confirm that extreme compression of the orthogonalized updates attains an accuracy comparable to that of Muon, while significantly reducing the volume of transmitted data.

For these experiments, we implemented a federated architecture with a central aggregation server. To keep the results comparable with the previous stages, we used the convolutional neural network (CNN2), comprising two convolutional blocks with BatchNorm and an MLP classifier. The CIFAR-10 dataset was partitioned across the clients homogeneously (homogeneous split). Within this learning type, we study both the baseline Federated SignMuon scheme (algorithm 2) and the error-feedback variant Federated EF-SignMuon (algorithm 3). Each communication round consists of a local gradient accumulation step on the clients, followed by transmitting the sign-compressed updates to the server, which updates the global model.

For a systematic analysis of the proposed algorithms, this experimental stage is split into several sub-experiments. First, we assess how the scale of the federated network affects the stability of sign-based optimization and the final accuracy of the model. To this end, we study the baseline SignMuon algorithm for different numbers of active clients.

Experiment 1: Scalability of SignMuon with respect to the number of clients. We investigate the effect of the number of participating clients on the training quality of SignMuon. The number of clients is $N \in \{1, 3, 5, 7, 9\}$, with 2000 training rounds and one local step per client. The results show that classification accuracy grows with the number of clients. This effect is explained by the improved quality of majority-vote aggregation as more independent sign estimates of the gradient become available. The results of this experiment are reported in Appendix A.9.

Having confirmed the positive effect of the number of clients on the stability of the algorithm, we proceed to a direct comparison of SignMuon with the existing algorithms. For this purpose, we select a configuration with 3 clients, which allows us to evaluate the efficiency of the algorithms under limited federation resources.

Experiment 2: Comparison of the algorithms with $N = 3$ clients. We compare 6 optimizers with 3 clients, 2000 rounds, and 1 local step. The results of this experiment are reported in Appendix A.10.

However, the potential of sign-based optimization algorithms is most fully revealed when the scale of the network and the amount of local computation increase. Therefore, at the final testing stage we increased the number of clients to 10.

Experiment 3: Comparison of the algorithms with $N = 10$ clients. The same 6 optimizers are tested with 10 clients, 2000 rounds, and 3 local steps per client. In this learning type, SignMuon reaches the quality of Muon. We conclude that, with a sufficient number of clients ($N \geq 10$), SignMuon achieves a quality comparable to Muon while reducing the volume of transmitted data by a factor of $\sim 32\times$, by transmitting only the signs of the gradients instead of their full values. The results of this experiment are reported in Table 4 and Figure 7.

Table 4: CIFAR-10: federated learning. Experiment 3: Comparison of the algorithms with $N = 10$ clients.

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	10	3	64	0.001	0.9	84.57%
	Muon					0.05		85.22%
	SignSGD					0.001		78.98%
	SGD					0.03		81.38%
	Adam					0.001		78.36%
	EF-SignMuon					0.02		84.59%

6 Conclusion

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A Appendix

A.1 Centralized algorithm

Algorithm 1 SignMuon

Input: Initial model $\mathbf{X}_0$, initial gradient $\mathbf{G}_0$, momentum coefficient μ, learning rate η_t, $\lambda_{\text{mult}} = 1$, number of Newton-Schulz iterations ns_steps

Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$

for $t = 0$ to $(T-1)$ **do**

  Stochastic gradient $\mathbf{G}_t$

  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + \mathbf{G}_t$ ▷ Momentum accumulation

  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov momentum),} \\ \mathbf{M}_t, & \text{(otherwise)} \end{cases}$

  $\mathbf{Y} \leftarrow \text{ReshapeTo2D}(\tilde{\mathbf{M}}_t)$ ▷ Flatten tensor to matrix $m \times n$

  for $k = 1$ to ns_steps **do** ▷ Newton-Schulz orthogonalization

    $\mathbf{A} \leftarrow \mathbf{Y}\mathbf{Y}^\top$

    $\mathbf{Y} \leftarrow a\,\mathbf{Y} - b\,\mathbf{A}\mathbf{Y} + c\,\mathbf{A}^2\mathbf{Y}$

  end for

  $\mathbf{D}_t \leftarrow \text{ReshapeToOriginal}(\mathbf{Y})$ ▷ Reshape back to original tensor

  $\mathbf{s}_t \leftarrow \text{sign}(\mathbf{D}_t)$ ▷ Sign compression

  $\mathbf{X}_{t+1} \leftarrow \mathbf{X}_t - \eta_t \mathbf{s}_t$ ▷ Update parameters

end for

A.2 Federated Algorithm

Algorithm 2 Federated SignMuon

Input: Initial model $\mathbf{X}_0$, clients number N, rounds number rounds, local steps number $\mathrm{n\_steps}$, learning rate for the main layers η_t, learning rate for the last layer η_t^{aux}, momentum coefficient μ
Output: Updated global model $\mathbf{X}$

for $j = 1$ to N **do**
  $\mathbf{M}_0^{(j)} \leftarrow 0$ ▷ Initialization of local buffers on clients
end for
Initialize AdamW for last_layer with learning rate η_0^{aux}
for $t = 0$ to $(\mathrm{rounds} - 1)$ **do**
  for $j = 1$ to N **in parallel do**
    $\mathbf{X}_t^{(j)} \leftarrow \mathbf{X}_t$ ▷ Retrieve global model
    for $k = 1$ to $\mathrm{n\_steps}$ **do**
      Sample a batch $\xi_{t,k}^{(j)}$
      $\nabla f_k \leftarrow \nabla f(\mathbf{X}_t^{(j)}; \xi_{t,k}^{(j)})$
      $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} + \nabla f_k$
    end for

    $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} / \mathrm{n\_steps}$ ▷ Gradient averaging
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$ ▷ Local momentum update
    $\mathbf{Y}_t^{(j)} \leftarrow \mathrm{ReshapeTo2D}(\mathbf{M}_t^{(j)})$ ▷ Reshape momentum to matrix $m \times n$

    for $k = 1$ to $\mathrm{ns\_steps}$ **do** ▷ Orthogonalization (Muon LMO)
      $\mathbf{A}_t^{(j)} \leftarrow \mathbf{Y}_t^{(j)} (\mathbf{Y}_t^{(j)})^\top$
      $\mathbf{Y}_t^{(j)} \leftarrow a\, \mathbf{Y}_t^{(j)} - b\, \mathbf{A}_t^{(j)} \mathbf{Y}_t^{(j)} + c\, (\mathbf{A}_t^{(j)})^2 \mathbf{Y}_t^{(j)}$
    end for

    $\mathbf{D}_t^{(j)} \leftarrow \mathrm{ReshapeToOriginal}(\mathbf{Y}_t^{(j)})$
    $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}\big(\mathbf{D}_t^{(j)}\big)$ ▷ Sign compression

    Send $\mathbf{s}_t^{(j)}$ **to server**
    for each $\mathbf{W}^{\mathrm{last}}$ **in last_layer do**
      Send $\mathbf{G}_t^{(j)}$ to the server ▷ Raw gradient for the head
    end for
  end for
  $\mathbf{s}_t^{\mathrm{agg}} \leftarrow \mathrm{sign}\Big(\sum_{j=1}^{N} \mathbf{s}_t^{(j)}\Big)$ ▷ Majority vote (matrix layers)
  $\bar{\mathbf{G}}_t^{\mathrm{last}} \leftarrow \frac{1}{N} \sum_{j=1}^{N} \mathbf{G}_t^{(j)}$ ▷ Average raw gradient for the head
  $\mathbf{X}_{t+1} \leftarrow \mathbf{X}_t - \eta_t \mathbf{s}_t^{\mathrm{agg}}$ ▷ Matrix parameter update

  $\mathbf{X}_{t+1}^{\mathrm{last}} \leftarrow \mathrm{AdamW}(\mathbf{X}_t^{\mathrm{last}}, \bar{\mathbf{G}}_t^{\mathrm{last}}, \eta_t^{\mathrm{aux}})$ ▷ AdamW on the averaged gradient
end for
return $\mathbf{X}_{\mathrm{rounds}}$

A.3 Federated Algorithm with Error Feedback

Algorithm 3 Federated EF-SignMuon

Input: Initial model $\mathbf{X}_0$, clients number N, rounds number rounds, local steps number $\mathrm{n\_steps}$, learning rate for the main layers η_t, learning rate for the last layer η_t^{aux}, momentum coefficient μ
Output: Updated global model $\mathbf{X}$

$\mathbf{G}_0 \leftarrow 0$ ▷ Initialize the global gradient estimate on the server (for the main layers)
for $j = 1$ to N **do**
$\quad$ $\mathbf{g}_0^{(j)} \leftarrow 0,\ \mathbf{M}_0^{(j)} \leftarrow 0$ ▷ Initialize local estimators and momentum
end for
Initialize AdamW for last_layer with learning rate η_0^{aux}
for $t = 0$ to $(\mathrm{rounds} - 1)$ **do**
$\quad$ **for** $j = 1$ to N **in parallel do** ▷ Local step of EF21
$\quad\quad$ $\mathbf{X}_t^{(j)} \leftarrow \mathbf{X}_t$ ▷ Retrieve global model
$\quad\quad$ $\mathbf{G}_t^{(j)} \leftarrow 0$
$\quad\quad$ **for** $k = 1$ to $\mathrm{n\_steps}$ **do**
$\quad\quad\quad$ Sample a batch $\xi_{t,k}^{(j)}$
$\quad\quad\quad$ $\nabla f_k \leftarrow \nabla f(\mathbf{X}_t^{(j)}; \xi_{t,k}^{(j)})$
$\quad\quad\quad$ $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} + \nabla f_k$
$\quad\quad$ **end for**
$\quad\quad$ $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} / \mathrm{n\_steps}$ ▷ Gradient averaging
$\quad\quad$ $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$ ▷ Local momentum update
$\quad\quad$ **for each** parameters $\mathbf{W}$ **not in last_layer do**
$\quad\quad\quad$ $\Delta_t^{(j)} \leftarrow \mathbf{M}_t^{(j)} - \mathbf{g}_t^{(j)}$ ▷ Difference computation
$\quad\quad\quad$ $\alpha_t^{(j)} \leftarrow \mathrm{mean}(|\Delta_t^{(j)}|)$ ▷ Adaptive scaling factor
$\quad\quad\quad$ $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}(\Delta_t^{(j)})$ ▷ Compress shifted gradient
$\quad\quad\quad$ $\mathbf{g}_{t+1}^{(j)} \leftarrow \mathbf{g}_t^{(j)} + \alpha_t^{(j)} \mathbf{s}_t^{(j)}$ ▷ Compute gradient estimator
$\quad\quad\quad$ Send $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$ to the server
$\quad\quad$ **end for**
$\quad\quad$ **for each** parameters $\mathbf{W}^{\mathrm{last}}$ **in last_layer do**
$\quad\quad\quad$ Send $\mathbf{G}_t^{(j)}$ to the server
$\quad\quad$ **end for**
$\quad$ **end for**
$\quad$ $\mathbf{G}_{t+1} \leftarrow \mathbf{G}_t + \frac{1}{N} \sum_{j=1}^{N} \alpha_t^{(j)} \mathbf{s}_t^{(j)}$ ▷ Update the global estimator on the server
$\quad$ **for each** parameters $\mathbf{X}$ **do**
$\quad\quad$ $\mathbf{Y} \leftarrow \mathrm{ReshapeTo2D}(\mathbf{G}_{t+1})$
$\quad\quad$ **for** $k = 1$ to $\mathrm{ns\_steps}$ **do** ▷ Orthogonalization (Muon LMO)
$\quad\quad\quad$ $\mathbf{A} \leftarrow \mathbf{Y}\mathbf{Y}^\top$
$\quad\quad\quad$ $\mathbf{Y} \leftarrow a\,\mathbf{Y} - b\,\mathbf{A}\mathbf{Y} + c\,\mathbf{A}^2\mathbf{Y}$
$\quad\quad$ **end for**
$\quad\quad$ $\mathbf{D}_{t+1} \leftarrow \mathrm{ReshapeToOriginal}(\mathbf{Y})$
$\quad\quad$ $\mathbf{X}_{t+1} \leftarrow \mathbf{X}_t - \eta_t \mathbf{D}_{t+1}$ ▷ Update the matrix parameters
$\quad$ **end for**
$\quad$ $\nabla^{\mathrm{last}} \leftarrow \frac{1}{N} \sum_{j=1}^{N} \mathbf{G}_t^{(j)}$
$\quad$ $\mathbf{X}_{t+1}^{\mathrm{last}} \leftarrow \mathrm{AdamW}(\mathbf{X}_t^{\mathrm{last}}, \nabla^{\mathrm{last}}, \eta_t^{\mathrm{aux}})$ ▷ Classifier update
end for
return $\mathbf{X}_{\mathrm{rounds}}$

A.4 Research Hypotheses

We formulate two main hypotheses that guide the theoretical and empirical investigation of the proposed algorithms. Let $\{\mathbf{X}_t\}_{t\geq 0}$ denote the sequence of global model parameters generated by the optimization algorithm, and $f(\mathbf{X})$ is the target function in federated learning setting (Equation 2).

To analyze the algorithms in both ways, we will make several assumptions.

Assumption 1 (Lower Boundedness). *The objective function $f(\mathbf{X})$ is bounded below by some value f^*, i.e., $f(\mathbf{X}) \geq f^*$ for all $\mathbf{X} \in \mathcal{X}$.*

Assumption 2 (Layer-wise Smoothness). *Following the layer-wise smoothness framework of Gruntkowska et al. Gruntkowska et al. (2025) (Assumptions 6–9), each matrix-valued parameter $\mathbf{X}_i \in \mathbb{R}^{m_i \times n_i}$, $i \in [p]$, is equipped with the spectral norm $\|\cdot\|_{2\to 2}$ and its dual, the nuclear norm $\|\cdot\|_*$. The global objective f and each local objective f_j, $j \in [N]$, are layer-wise L-smooth: for every layer $i \in [p]$ and all $\mathbf{X}, \mathbf{Y} \in \mathcal{X}$,*

$$\|\nabla_i f(\mathbf{X}) - \nabla_i f(\mathbf{Y})\|_* \leq L_i \,\|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to 2}, \qquad \|\nabla_i f_j(\mathbf{X}) - \nabla_i f_j(\mathbf{Y})\|_* \leq L_{i,j} \,\|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to 2}, \tag{13}$$

with layer constants $L_i, L_{i,j} \geq 0$; we denote $L := \max_{i\in[p]} L_i$. For a single matrix layer this reduces to $\|\nabla f(\mathbf{X}) - \nabla f(\mathbf{Y})\|_* \leq L\,\|\mathbf{X} - \mathbf{Y}\|_{2\to 2}$. The layer-wise (L_0, L_1)-smoothness variant of Gruntkowska et al. (2025) (Assumptions 8–9) is recovered by replacing L_i (resp. $L_{i,j}$) with $L_{0,i} + L_{1,i}\|\nabla_i f(\mathbf{X})\|_*$ (resp. $L_{0,i,j} + L_{1,i,j}\|\nabla_i f_j(\mathbf{X})\|_*$).

Assumption 3 (Stochastic Gradient). *The stochastic gradient at each client $j \in [N]$ is unbiased and has bounded variance in the dual norm:*

- *Unbiased:*
$$\mathbb{E}_{\xi\sim\mathcal{D}_j}\big[\nabla f_j(\mathbf{X};\xi)\big] = \nabla f_j(\mathbf{X}), \tag{14}$$
- *Bounded variance:*
$$\mathbb{E}_{\xi\sim\mathcal{D}_j}\big[\|\nabla f_j(\mathbf{X};\xi) - \nabla f_j(\mathbf{X})\|_*^2\big] \leq \sigma^2, \quad \sigma > 0. \tag{15}$$

Assumption 4 (Biased Compression). *We impose conditions on the compressor, following Beznosikov et al. (2024). There exist constants $\gamma > 0$ and $\beta > 0$ such that for any matrix $\mathbf{Y} \in \mathbb{R}^{m\times n}$, the sign operator $\mathcal{C}(\mathbf{Y}) = \mathrm{sign}(\mathbf{Y})$ satisfies*

$$\gamma\|\mathbf{Y}\|_1 \leq \langle \mathbf{Y}, \mathrm{sign}(\mathbf{Y})\rangle = \|\mathbf{Y}\|_1, \tag{16}$$
$$\|\mathrm{sign}(\mathbf{Y})\|_F^2 \leq \beta\|\mathbf{Y}\|_1. \tag{17}$$

Hypothesis 1 (Convergence). *Under Assumptions 1–4, the Federated SignMuon algorithm converges in the smooth non-convex setting. There exists a choice of the learning rate η_t such that the gradient norm satisfies the bound*

$$\min_{0\leq t\leq T} \mathbb{E}\big[\|\nabla f(\mathbf{X}_t)\|_*^2\big] = \mathcal{O}\big(T^{-1/2}\big). \tag{18}$$

This means that applying extreme 1-bit sign compression after matrix orthogonalization (LMO) does not worsen the asymptotic convergence rate compared to full-precision non-Euclidean methods in the smooth non-convex case.

Hypothesis 2 (Preservation of Muon's Advantages). *Replacing the Muon LMO direction $\mathbf{D}_t = \mathbf{U}_t\mathbf{V}_t^\top$ by its sign compression $\mathrm{sign}(\mathbf{D}_t)$ preserves the core geometry of orthonormal updates. As a result, SignMuon is expected to retain training quality close to that of full Muon at a substantially lower communication cost, and to outperform SignSGD in both the centralized and the federated regimes.*

A.5 Convergence Bounds for EF-SignMuon

Our implementation of Federated EF-SignMuon (Algorithm A.3) uses the following steps: the server broadcasts the full model $\mathbf{X}_t$, which is equivalent to choosing the server compressor $C^k \equiv I$ in the framework of Gruntkowska et al. (2025). Compression is applied only on the client side, through the scaled-sign operator. This is the regime analyzed in Lemmas 7–8 of Gruntkowska et al. (2025); below we restate the two resulting bounds in our notation. For the analysis we adopt the exponential moving-average (EMA) momentum convention used in Gruntkowska et al. (2025):

$$\mathbf{M}_t^{(j)} = (1-\beta)\,\mathbf{M}_{t-1}^{(j)} + \beta\,\nabla f_j(\mathbf{X}_t;\xi_t^{(j)}), \qquad \beta \in (0,1].$$

It is equivalent to the heavy-ball form $\mathbf{M}_t^{(j)} = \mu\mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$ of Algorithm A.3 up to a rescaling of the momentum buffer.

Assumption 5 (Contractive Compression). *The scaled-sign compressor* $\mathcal{C}(\mathbf{Y}) = \mathrm{mean}|\mathbf{Y}| \cdot \mathrm{sign}(\mathbf{Y}) = \frac{1}{d}\|\mathbf{Y}\|_1\,\mathrm{sign}(\mathbf{Y})$*, where* $\mathbf{Y} \in \mathbb{R}^{m\times n}$ *and* $d = mn$*, is contractive:*

$$\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 \le (1-\alpha)\|\mathbf{Y}\|_F^2 \qquad \alpha \in (0,1], \qquad \forall\,\mathbf{Y} \in \mathbb{R}^{m\times n}. \tag{19}$$

For the scaled-sign operator one may take $\alpha = 1/d$*, since a direct computation gives* $\|\mathcal{C}(\mathbf{Y})-\mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - \frac{1}{d}\|\mathbf{Y}\|_1^2$ *and* $\|\mathbf{Y}\|_1^2 \ge \|\mathbf{Y}\|_F^2$*.*

Lemma 1 (Gradient-estimator contraction). *Let Assumptions 1–3 and 5 hold, and run EF-SignMuon with identity compression on the server. Then the compressed gradient estimator* $\mathbf{g}_t^{(j)}$ *of each client* $j \in [N]$ *satisfies*

$$\begin{aligned}\mathbb{E}\Big[\|\mathbf{M}_t^{(j)} - \mathbf{g}_t^{(j)}\|_F^2 \,\big|\, \mathbf{X}_t, \mathbf{M}_{t-1}^{(j)}, \mathbf{g}_{t-1}^{(j)}\Big] \\ \le (1-\alpha)\Big[(1-\beta)\,\|\mathbf{M}_{t-1}^{(j)} - \mathbf{g}_{t-1}^{(j)}\|_F^2 + 2\beta\,\|\nabla f_j(\mathbf{X}_t) - \mathbf{g}_{t-1}^{(j)}\|_F^2 + 2\beta\,\sigma^2\Big].\end{aligned} \tag{20}$$

This is Lemma 7 of Gruntkowska et al. (2025) specialized to the identity server regime ($C^k \equiv I$). The estimator $\mathbf{g}_t^{(j)}$ is contracted towards the momentum $\mathbf{M}_t^{(j)}$ at a geometric rate governed by the contraction α; the identity $\mathbf{M}_t^{(j)} - \mathbf{g}_t^{(j)} = (\mathbf{M}_t^{(j)} - \mathbf{g}_{t-1}^{(j)}) - \mathcal{C}(\mathbf{M}_t^{(j)} - \mathbf{g}_{t-1}^{(j)})$ reduces the bound to equation 19 together with the convexity of $\|\cdot\|_F^2$.

Lemma 2 (Momentum tracking). *Let Assumptions 1–3 and 5 hold, and let each* f_j *be* L*-smooth (Assumption 2). Then the EMA momentum tracks the true local gradient:*

$$\mathbb{E}\|\mathbf{M}_t^{(j)} - \nabla f_j(\mathbf{X}_t)\|_F^2 \le (1-\beta)\,\mathbb{E}\|\mathbf{M}_{t-1}^{(j)} - \nabla f_j(\mathbf{X}_{t-1})\|_F^2 + \beta\,\sigma^2 + C(\beta)\,L^2\,\|\mathbf{X}_t - \mathbf{X}_{t-1}\|_F^2, \tag{21}$$

with $C(\beta) = (1-\beta)/\beta$ *as in Lemma 8 of Gruntkowska et al. (2025).*

This is Lemma 8 of Gruntkowska et al. (2025) (identity server regime); the last term reflects the smoothness-induced drift and is proportional to the squared length of the LMO step $\mathbf{X}_t - \mathbf{X}_{t-1}$.

Lemmas 1-2 imply that the global estimator $\mathbf{G}_t = \frac{1}{N}\sum_{j=1}^N \mathbf{g}_t^{(j)}$ remains a controlled-bias approximation of the full gradient $\nabla f(\mathbf{X}_t)$. Combined with Assumption 2, this is the key ingredient behind the $\mathcal{O}(T^{-1/2})$ rate postulated in Hypothesis 1 for our variant of EF-SignMuon.

A.6 Proof of Theorem 1

Proof The proof is constructed using a counterexample that shows the existence of a smooth function and a matrix dimension $N \geq 4$ for which the SignMuon algorithm is guaranteed to diverge. We conjecture, supported by extensive numerical search, that no such counterexample exists for $N \leq 3$. However, at $N = 4$, this convergence guarantee vanishes.

Consider a simple linear objective function $f(\mathbf{W}) = \mathrm{Tr}(\mathbf{G}^\top \mathbf{W})$. For standard gradient descent or Muon, such a function diverges towards $-\infty$. The gradient is globally constant: $\nabla f(\mathbf{W}_t) = \mathbf{G}$ for all t. The change in the objective function at each iteration of the SignMuon algorithm is given by:

$$f(\mathbf{W}_{t+1}) - f(\mathbf{W}_t) = -\eta \langle \mathbf{G}, \mathrm{sign}(\mathbf{U}\mathbf{V}^\top) \rangle. \tag{22}$$

To prove divergence, it is sufficient to present a matrix $\mathbf{G}$ for which $\langle \mathbf{G}, \mathrm{sign}(\mathbf{U}\mathbf{V}^\top)\rangle < 0$. In this case, $f(\mathbf{W})$ strictly increases at every step, and the algorithm perpetually moves in the wrong direction.

We construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ by defining a specific orthogonal matrix $\mathbf{O}$ and a specific rank-1 principal component $\mathbf{u}_1\mathbf{v}_1^\top$. Let the orthogonal matrix $\mathbf{O}$ be given by the following exact rational numbers:

$$\mathbf{O} = \frac{1}{103}\begin{pmatrix} 101 & 20 & 2 & -2 \\ -20 & 97 & 20 & -20 \\ -2 & 20 & 2 & 101 \\ -2 & 20 & -101 & -2 \end{pmatrix}. \tag{23}$$

Because no entry is zero, its element-wise sign matrix $\mathbf{S} = \mathrm{sign}(\mathbf{O})$ is uniquely defined. Now, let $\mathbf{u}_1$ and $\mathbf{v}_1$ be the following exact unit vectors:

$$\mathbf{u}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix} 10 \\ -3 \\ 10 \\ 10 \end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix} 10 \\ 3 \\ -10 \\ 10 \end{pmatrix}. \tag{24}$$

One can easily verify that $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, meaning $\mathbf{u}_1$ and $\mathbf{v}_1$ act perfectly as left and right singular vectors for this orthogonal space. The crucial feature of this geometry is that the Frobenius inner product between this rank-1 component and the sign matrix $\mathbf{S}$ yields an exact, strictly negative fraction:

$$\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle = \mathbf{u}_1^\top \mathbf{S}\mathbf{v}_1 = -\frac{43}{103}. \tag{25}$$

We construct the gradient matrix $\mathbf{G}$ by assigning a massive singular value ($\sigma_1 = 1001$) to this pathological component and a singular value of 1 to the rest of the orthogonal space:

$$\mathbf{G} = \begin{pmatrix} 324.6 & 97.3 & -323.6 & 323.6 \\ -97.3 & -28.2 & 97.3 & -97.3 \\ 323.6 & 97.3 & -323.6 & 324.6 \\ 323.6 & 97.3 & -324.6 & 323.6 \end{pmatrix}. \tag{26}$$

Indeed, since $\mathbf{G} = 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}$ and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$,

$$\mathbf{G}\mathbf{v}_1 = 1000\,\mathbf{u}_1(\mathbf{v}_1^\top\mathbf{v}_1) + \mathbf{O}\mathbf{v}_1 = 1001\,\mathbf{u}_1,$$

so $(\mathbf{u}_1, \mathbf{v}_1)$ is a singular pair of $\mathbf{G}$ with $\sigma_1 = 1001$. For any $\mathbf{w} \perp \mathbf{v}_1$ we have $\mathbf{G}\mathbf{w} = \mathbf{O}\mathbf{w}$; as $\mathbf{O}$ is orthogonal and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, the restriction $\mathbf{O}|_{\mathbf{v}_1^\perp}$ is an isometry onto $\mathbf{u}_1^\perp$, hence $\sigma_2 = \sigma_3 = \sigma_4 = 1$. Moreover, with $\mathbf{u}_k = \mathbf{O}\mathbf{v}_k$ for $k \geq 2$,

$$\textstyle\sum_{k\geq 2} \mathbf{u}_k\mathbf{v}_k^\top = \mathbf{O}\big(\mathbf{I} - \mathbf{v}_1\mathbf{v}_1^\top\big) = \mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top,$$

so $\mathbf{U}_G\mathbf{V}_G^\top = \mathbf{u}_1\mathbf{v}_1^\top + (\mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top) = \mathbf{O}$.

Substituting the resulting matrix $\mathbf{G}$ into the descent condition yields:

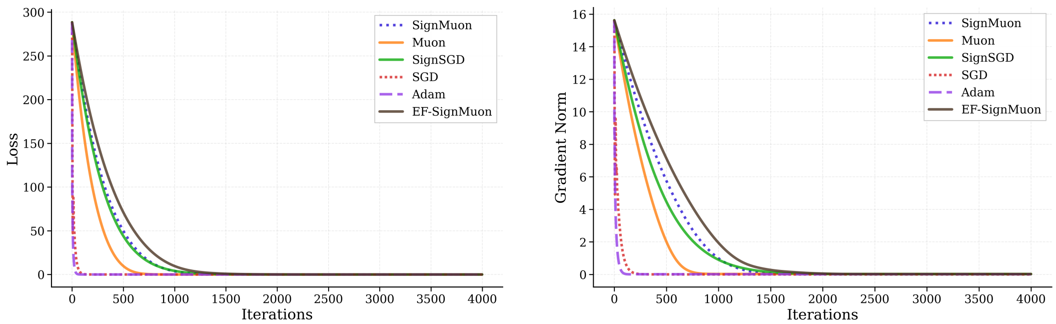
$$\langle \mathbf{G}, \mathbf{S}\rangle = \langle 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}, \mathbf{S}\rangle = 1000\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle + \langle \mathbf{O}, \mathbf{S}\rangle. \tag{27}$$

The inner product $\langle \mathbf{O}, \mathbf{S}\rangle$ is equivalent to the L_1 norm (sum of absolute values) of $\mathbf{O}$, which equals exactly $\frac{532}{103}$. Therefore:

$$\langle \mathbf{G}, \mathbf{S}\rangle = 1000\left(-\frac{43}{103}\right) + \frac{532}{103} = -\frac{42468}{103} \approx -412.31. \tag{28}$$

Because $\langle \mathbf{G}, \mathbf{S}\rangle < 0$, the change in the objective function at every iteration is given by $f(\mathbf{W}_{t+1}) = f(\mathbf{W}_t) + 412.31\eta$. Thus, instead of minimizing, the algorithm strictly and monotonically diverges to $+\infty$, which completes the proof. ■

A.7 Results for the smooth convex problem

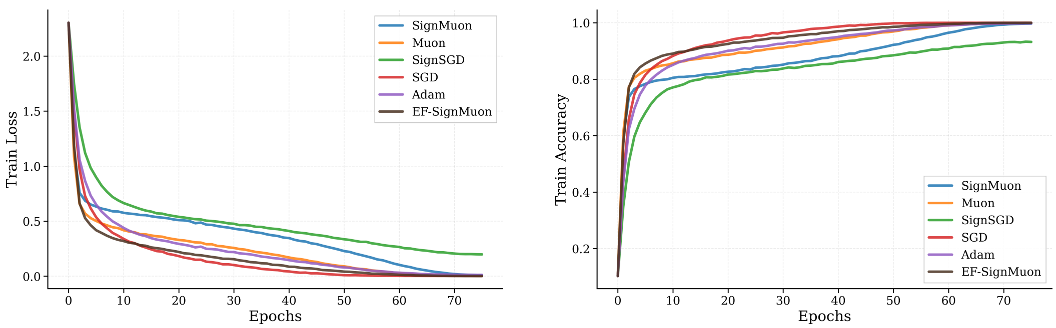


(a) Loss

(b) Frobenius norm of the gradient

Figure 2: Smooth convex problem (Equation 11 for 500×500).

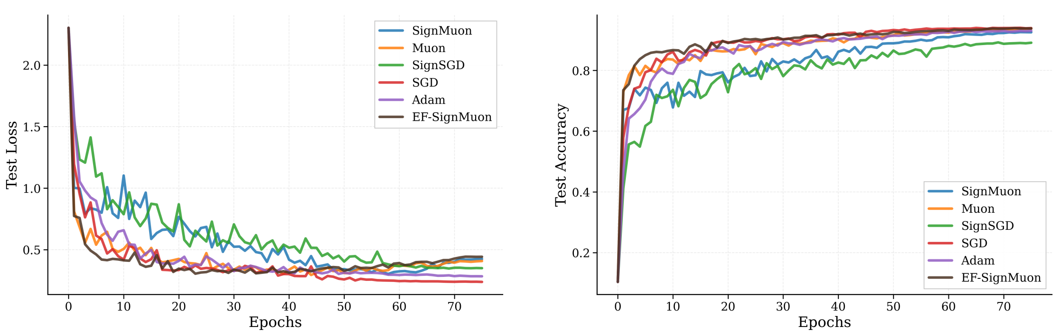
A.8 Centralized training results



(a) Train Loss

(b) Train Accuracy

Figure 3: Centralized setting. Comparison of convergence and classification accuracy achieved by different optimization algorithms during CIFAR-10 classifier training on the training set.



(a) Test Loss

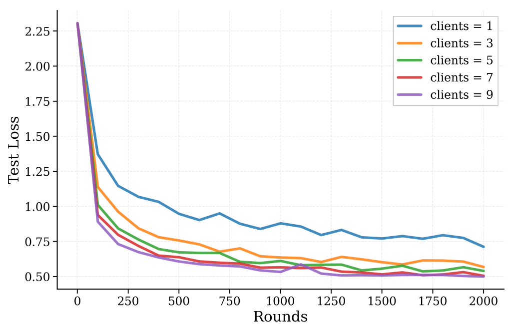
(b) Test Accuracy

Figure 4: Centralized setting. Comparison of convergence and classification accuracy achieved by different optimization algorithms during CIFAR-10 classifier training on the test set.

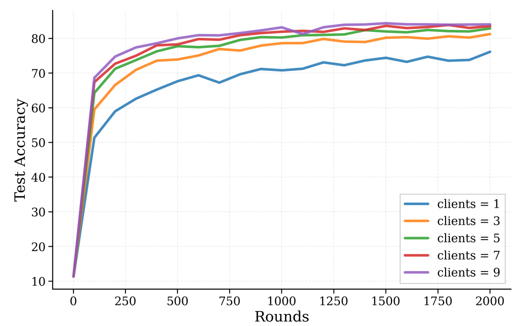
A.9 Federated training results. Experiment 1

Table 5: CIFAR-10 Federated Learning. Experiment 1. Scalability of the SignMuon Algorithm.

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	1	1	64	0.001	0.9	76.13%
			3					81.24%
			5					82.87%
			7					83.54%
			9					84.03%



(a) Test Loss



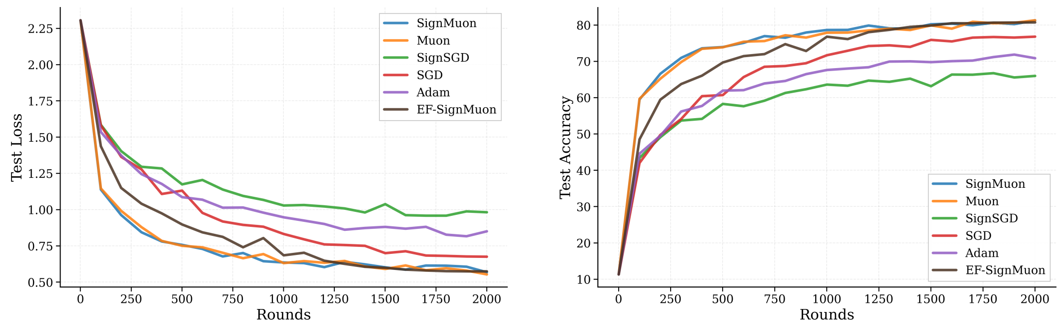
(b) Test Accuracy

Figure 5: CIFAR-10 Federated Learning. Experiment 1. Scalability of the SignMuon Algorithm.

A.10 Federated training results. Experiment 2

Table 6: CIFAR-10 Federated Learning. Experiment 2: Comparison of Optimization Algorithms with (N = 3) Clients.

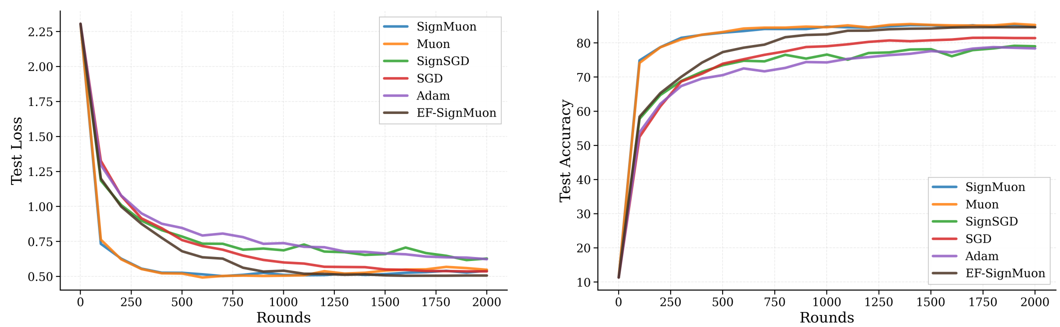
Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	3	1	64	0.001	0.9	81.24%
	Muon					0.05		81.27%
	SignSGD					0.001		65.96%
	SGD					0.03		76.77%
	Adam					0.001		70.83%
	EF-SignMuon					0.02		80.69%



(a) Test Loss (b) Test Accuracy

Figure 6: CIFAR-10 Federated Learning. Experiment 2: Comparison of Optimization Algorithms with (N = 3) Clients.

A.11 Federated training results. Experiment 3



(a) Test Loss (b) Test Accuracy

Figure 7: CIFAR-10 Federated Learning. Experiment 3: Comparison of Optimization Algorithms with (N = 10) Clients.